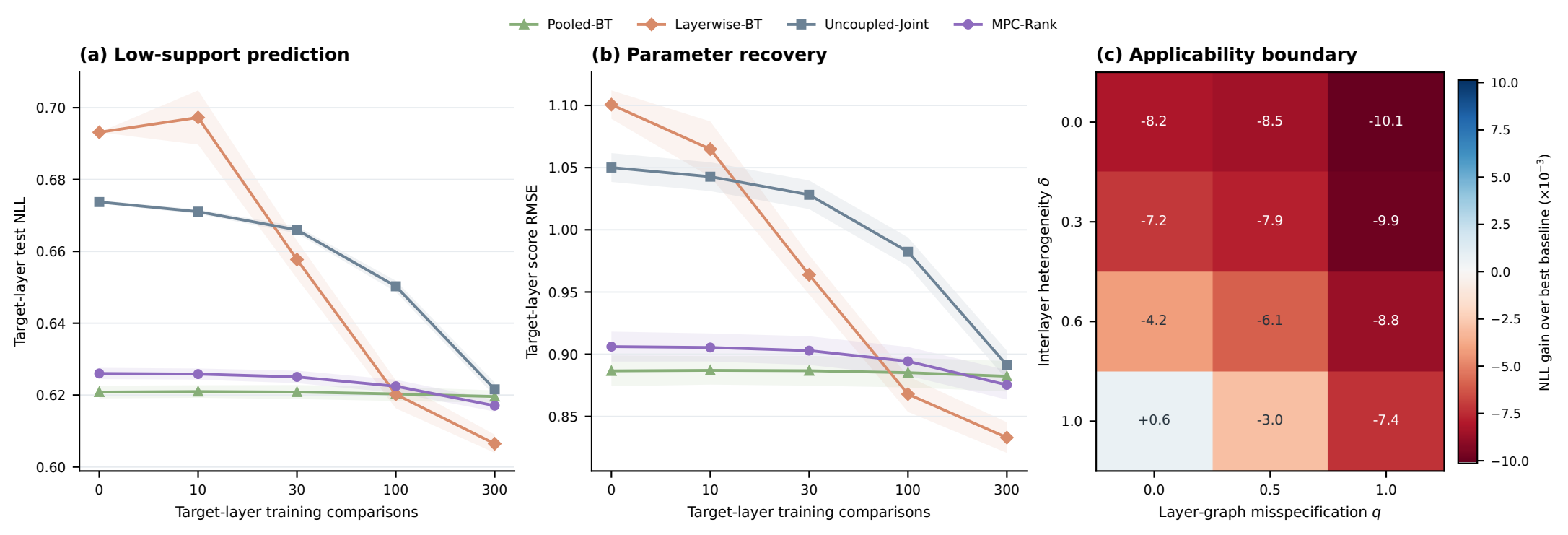
Pooled-BT
Layerwise-BT
Uncoupled-Joint
MPC-Rank
(a) Low-support prediction
Target-layer test NLL
0.70
0.68
0.66
0.64
0.62
0.60
0
10
30
100
300
Target-layer training comparisons
(b) Parameter recovery
Target-layer score RMSE
1.10
1.05
1.00
0.95
0.90
0.85
0
10
30
100
300
Target-layer training comparisons
(c) Applicability boundary
Interlayer heterogeneity δ
0.0
0.3
0.6
1.0
-8.2
-8.5
-10.1
-7.2
-7.9
-9.9
-4.2
-6.1
-8.8
+0.6
-3.0
-7.4
0.0
0.5
1.0
Layer-graph misspecification q
NLL gain over best baseline (×10⁻³)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0